

YRF STATA Exam (Quantitative stream)

Time: 9:00 AM to 3:00 PM

For the final test, you will use the STAR+ dataset. This dataset originates from a randomized controlled trial (RCT) in which participants, youths with disabilities, were assigned unique identifiers as the first-line numbering during the survey. The survey was stratified at the division level and randomized at the branch level.

1. Using the baseline data, create a long dataset for the demographic module. Follow the example dataset below and only keep the variables mentioned in the table-

Final id	Member id	Name	Sex	Age	Marital status
1	1	A	Female	28	Married
1	2	B	Male	32	Married
1	3	C	Female	7	Unmarried
2	1	D	Male	24	Unmarried
2	2	E	Female	32	Unmarried
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2. Using the baseline data, create a bar graph showing the distribution of different types of disabilities for treatment and control groups. Ensure proper labeling of axes and groups
3. Using the baseline data, create a balance table comparing treatment and control groups for the following variables: youth's age, youth's gender, youth's marital status, household size, amount of land, number of cows owned, number of goats or sheep owned, household monthly average income
4. Run a regression to test whether attrition is balanced between the treatment and control groups, controlling for youth's age, youth's gender, youth's marital status, household size, amount of land, number of cows owned, number of goats or sheep owned, household monthly average income, and their interaction with treatment status
5. Estimate the Intention-to-Treat (ITT) and Local Average Treatment Effect (LATE) for (1) youth's income and (2) youth's employment at endline. Save the resulting regression output into an Excel file.

Note: Use the treatment (*treatment*) assignment variable from the baseline dataset and the treatment uptake (*participation*) variable from the endline dataset.

6. Re-run the analyses from Question 4, adding baseline covariates as controls (use the variables generated for balancing tests as covariates). Append these results to the Excel file generated for Question 4
7. Explore the treatment effect on male and female participants controlling for baseline covariates
8. Create a balanced panel dataset in long format. Use the panel to:

(a) Run a Difference-in-Differences analysis to estimate the treatment effect on income (without baseline covariates).

(b) Run a fixed-effects model for the same outcome.

Compare and report the results in an Excel file